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Effects of Pesticides on Birds Egg

DDT is a pesticide that was widely used on crops until the 1970s. DDT was banned in several countries because of the harm it was causing to birds. DDT causes some birds to produce thin egg shells, decreasing the survival rates of birds' offsprings. This resulted in decreasing population of several species. The thinning of the eggs was so significant that even the weight of an incubating bird could crush the eggs. With the ban of DDT in some countries, populations of the birds have increased.

Urinogenital system

Excretory System

The paired kidneys which are **metanephric** are flat, elongated and lobulated. The ureters lead directly backward to open into the **urodaeum** or middle compartment of the cloaca; there is no urinary bladder. The nitrogenenous waste is excreted in the form of uric acid and discharged as a semi-solid mass. **Adrenal bodies** lie attached to the ventral surface of the kidneys as small yellowish elongated streaks.

Reproductive system.

A pair of ovoid **testes** are attached to anterior end of the kidneys by peritoneum (Figure 4.36). From each testis leads the **vas deferens** which runs backwards along the outer side of the ureter of that side, and opens on a small papilla into the urodaeum. The vas deferens is dilated into a **seminal vesicle** at its hind end. There is no copulatory organ.

The **female reproductive organs** consist of a single **ovary** on the left side which is an **adaptation** to aerial life and an oviduct which opens into the body-cavity by a funnel-like aperture at the anterior end and posteriorly opens into the **urodaeum** (Figure 4.37).

Pigeons can be trained to distinguish identities and emotional expressions on human faces, letters of the alphabet, misshapen pharmaceutical capsules, and even paintings. Pigeons are sensitive to diagnostically salient features in medical images such as slides and x-rays to diagnose and classify disease risk. When trained, pigeons can correctly identify, distinguish normal and cancerous digitized slides and X rays. Pigeons are remarkably adept at discriminating between benign and malignant breast cancer slides at all magnification.

Activity 1

1 Examine a quill feather. Hold the base of the central shaft with one hand and gently bend the tip of the feather with your other hand. Be careful not to break the feather. Next, hold the shaft and wave the feather in the air. Record your observations concerning the structure of the quill feather. Relate your observations to the feather's possible function. Describe the function of the feather under "Function of feather" in your data table.

2 Examine the vane of the feather. Does the vane appear to be a solid structure? Include a description of the quill feather's vane structure under "Structure of feather" in your data table.

3 Make a drawing of the quill feather. Label the shaft, vanes and barbs. Compare your diagram with the feather.